KSE 2025

Da Lat, Vietnam

AI-THENA

TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN ĐHQG-HCM

fit@hcmus
VNUHCM - UNIVERSITY OF SCIENCE
FACULTY OF INFORMATION TECHNOLOGY

Flame Reavers@ALQAC 2025: Integrating Learned Rankers and LLM Reasoning in a Dynamic Hybrid Architecture for Legal Retrieval

Huy Trieu, Dang Phuong Nam Doan, Anh Kiet Nguyen, Thanh Thai Nguyen, Thanh Nghia Vo, Tung Le and Huy Tien Nguyen

Presenter: Dang Phuong Nam Doan

Email: ddpnam22@clc.fitus.edu.vn

Content

1. Introduction

Introduction

- **Task 1: Legal Document Retrieval**
  - Retrieve all legal articles relevant to a given question.
  - Metric: **Macro F2-score** (focus on recall).
- **Task 2: Legal Question Answering**
  - Answer the question based on retrieved articles.
  - Formats: **True/False**, **Multiple Choice**, **Free-text**.
  - Metric: **Accuracy**.

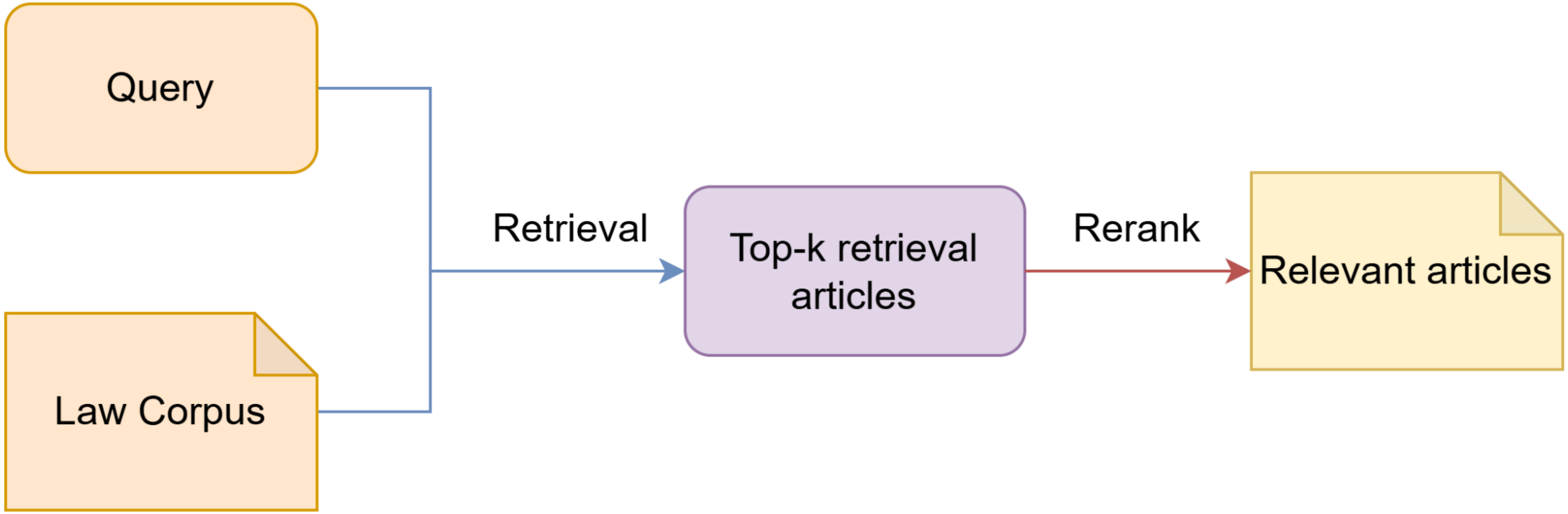
2. Related Works

Related Works

- **Task 1: Legal Document Retrieval**
  - From **lexical baselines** (BM25, TF-IDF) to neural **retrieve-and-rerank** pipelines (BM25 + BERT).
  - Recent works integrate **LLMs** for summarization or reranking (CAPTAIN from COLIEE 2024).
- **Task 2: Legal Question Answering**
  - Evolved from rule-based to **BERT-based** models (DeBERTa, RoBERTa).
  - Enhanced via **augmentation**, **entity logic**, and **span extraction** (KIS, HUKB).

3. Methodology

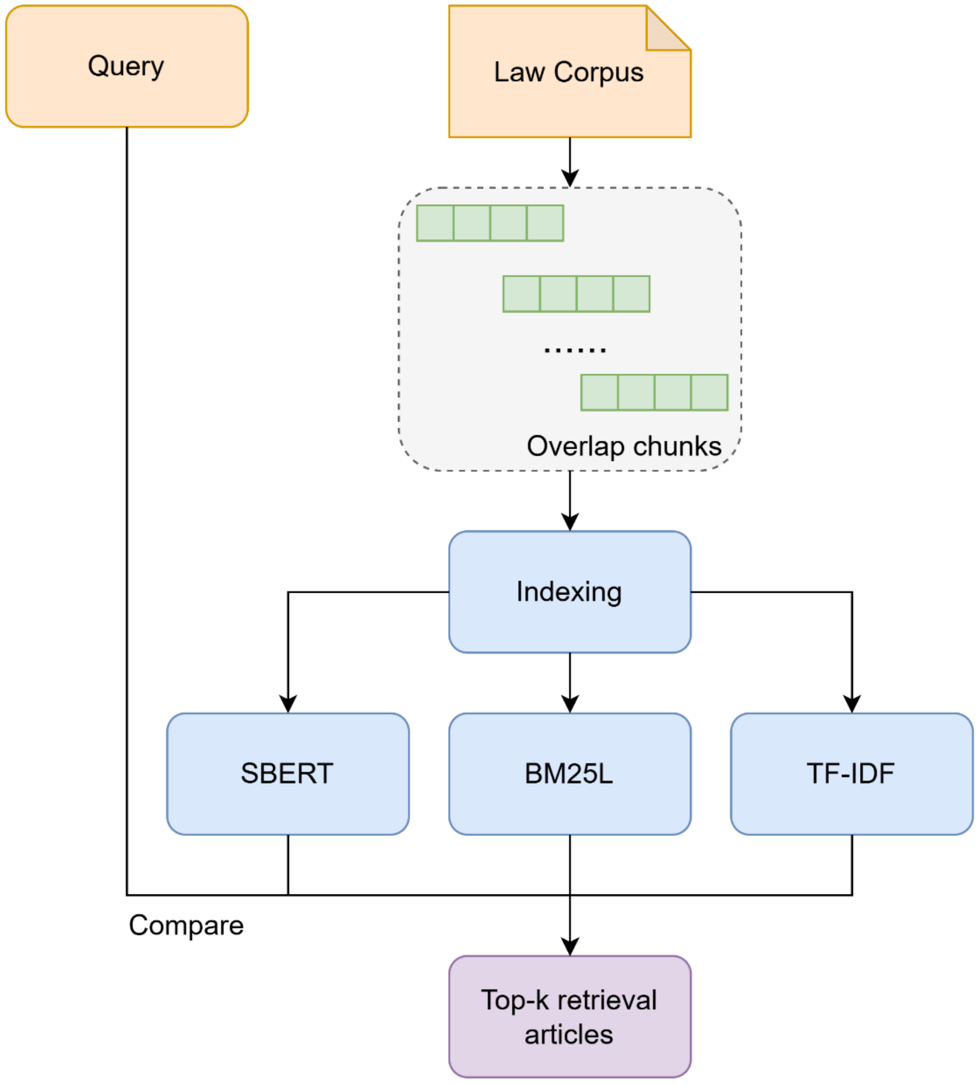
Task 1: Legal Document Retrieval



A Common Pipeline for Legal Document Retrieval

Hybrid Retrieval

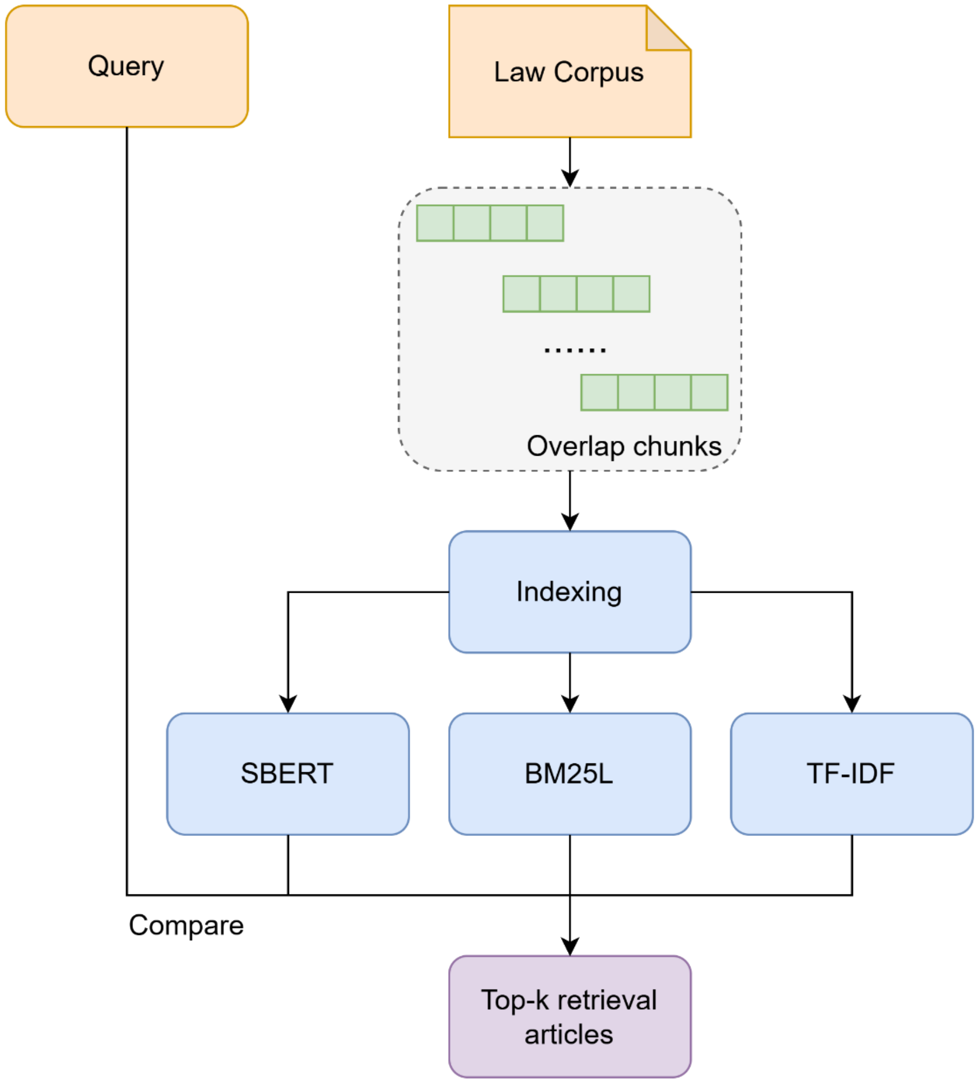
- **Preprocessing**: Document chunking with overlapping chunks.
- **Components:**
  - **Sparse Models:** BM25L & TF-IDF (for keyword matching).
  - **Dense Model:** Vietnamese Embeddings (for semantic understanding).
- **Fusion Method:**
  - Score = α * BM25L + β * TF-IDF + γ * Embedding.



Hybrid Retrieval Weight Optimization

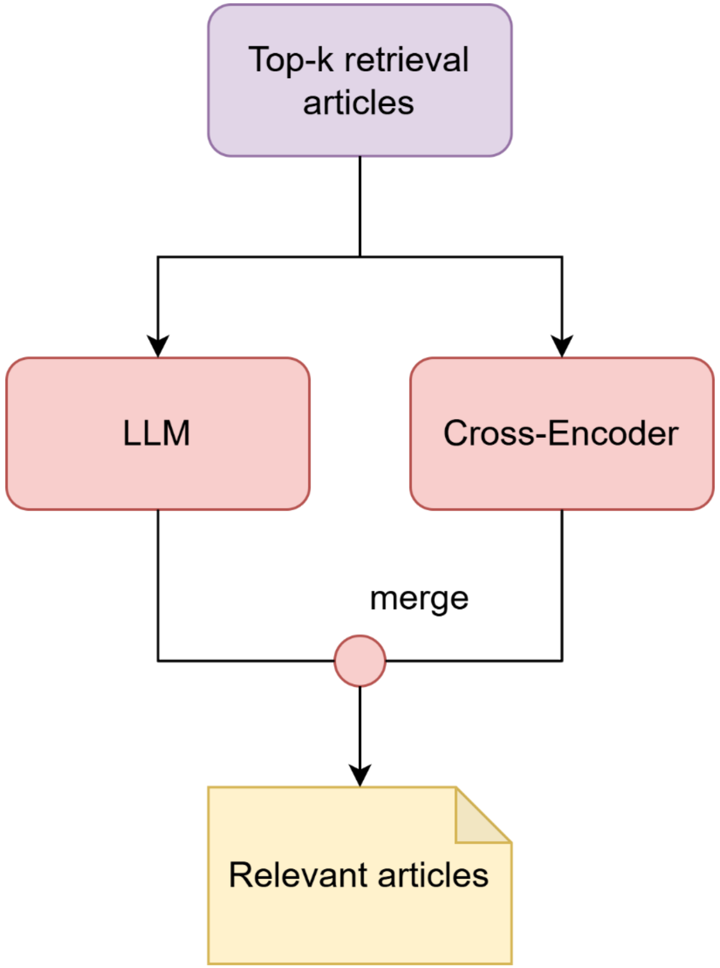
fit@hcmus

- **Goal**: Find weights (α, β, γ) such that
  - For each query: **Score(relevant) > Score(non-relevant) (1).**
- **Method**
  - Perform **grid search** over α, β.
  - Set $\boldsymbol{\gamma = 1 - \alpha - \beta}$.
- **Selection**
  - Choose (α, β, γ) with the **highest number of queries** satisfying **(1)**.



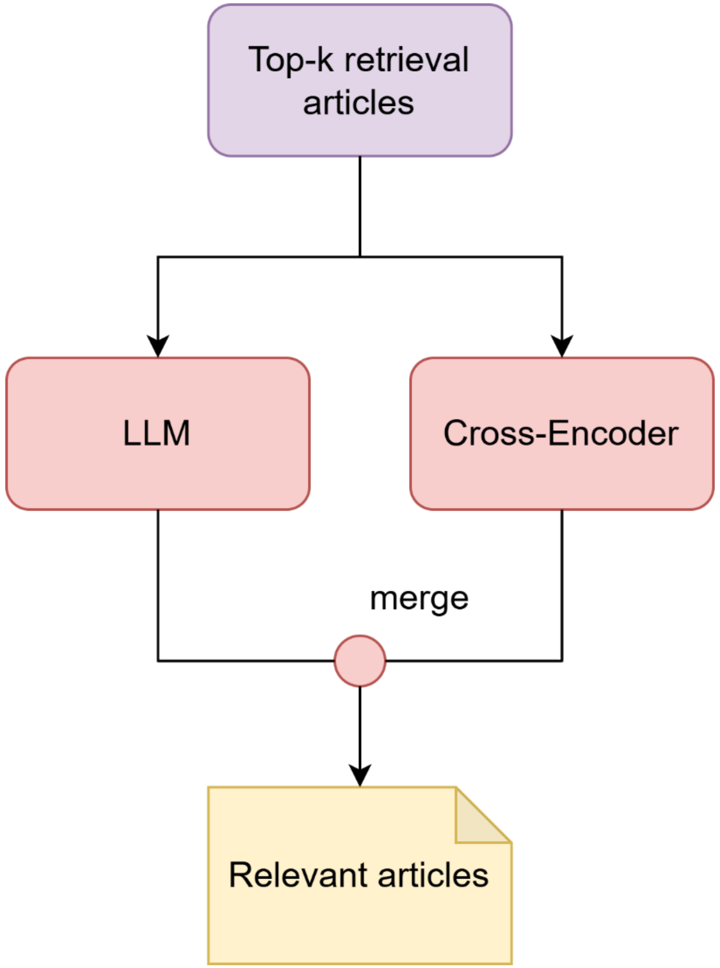
Hybrid Reranking Strategy

- **Our Approach:** A two-pronged strategy featuring parallel rerankers:
  - A fine-tuned Cross-Encoder.
  - An LLM-based Reasoning Module.



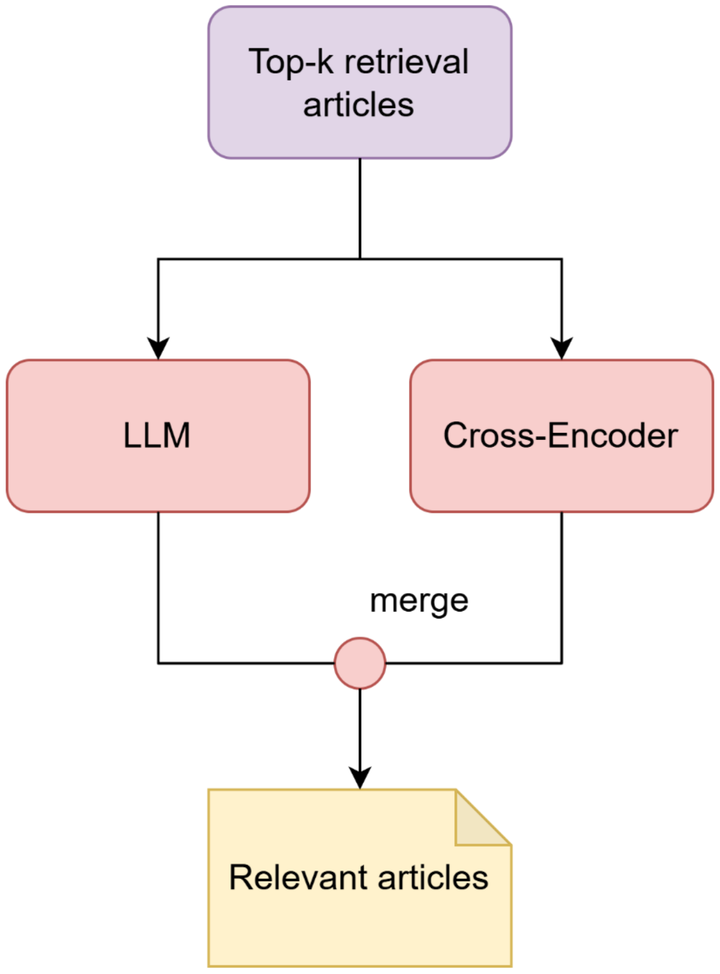
Fine-tuning Phase

- **Data preparation**:
  - Positive = relevant articles.
  - Hard negatives = top-20 non-relevant with **highest similarity**.
- **Model training**:
  - Fine-tune reranker as **Cross-Encoder**.



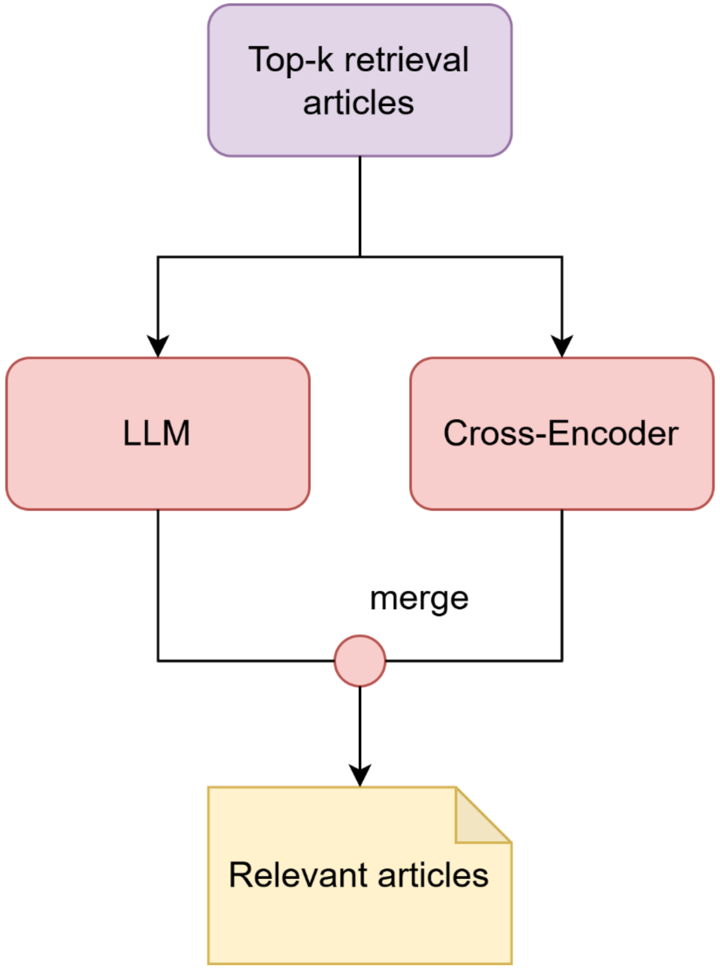
Inference Phase

- Retrieve **top-k** articles from Hybrid Retrieval.
- Rerank with fine-tuned Cross-Encoder.
- **Decision rule**:
  - If any article score > **0.8** → select those articles.
  - Else → use **LLM reranker**.



LLM Reranking Strategy

- Each article is classified into **one of four levels**:
  - **Very Relevant**: Directly and clearly answers the query.
  - **Relevant**: Addresses the query but partially.
  - **Less Relevant**: Indirect or limited connection to the query.
  - **Not Relevant**: Unrelated to the user's legal concern.
- **Selection Rule**:
  - Prioritize "**Very Relevant**" documents → if none, choose "**Relevant**", then "**Less Relevant**", following this hierarchy.



Task 2: Legal Question Answering

fit@hcmus

- **Method**: Tailored Prompting Strategy.
  - **True/False**: Output only "True" or "False".
  - **Multiple Choice**: Output only one letter (A/B/C/D).
  - **Free-text**: Generate concise answers grounded in provided articles.
- **Core Principle:** Model reasoning is strictly limited to the given legal text to minimize hallucination and ensure factual accuracy.
- **Selected Model**: Qwen/Qwen3-8B

4. Experiments and Results

Evaluation Metrics (Task 1)

- **Primary Metric: F2-macro**, emphasizing Recall to avoid missing relevant articles:

$$\text{F2-macro} = \frac{1}{|Q|}\sum_{i=1}^{|Q|}\left(\frac{5 \times \text{Precision}_i \times \text{Recall}_i}{4 \times \text{Precision}_i + \text{Recall}_i}\right).$$

- For a more comprehensive analysis of ranking quality, we also report **mAP** (Mean Average Precision) and **R-Precision**:

$$\text{mAP} = \frac{1}{|Q|}\sum_{q\in Q}\text{AP}_q,$$

$$\text{R-Precision}_i = \frac{r_i}{R_i}.$$

Evaluation Metrics (Task 2)

fit@hcmus

- **Primary Metric: Accuracy**, measures the percentage of correctly answered questions.

$$\text{Accuracy} = \frac{(\text{number of correct predictions})}{(\text{total number of instances})}$$

- **Evaluation Notes**:
  - The system is provided with ground-truth relevant articles as context.
  - Answers to free-text (essay) questions are manually evaluated by experts.

Evaluation Results (Task 1)

fit@hcmus

Teams	Methods	F2-macro	mAP	R-Precision
Ours	**Our fine-tuned - LLM reranker**	**0.9435**	**0.9512**	**0.9329**
-	AITeamVN - LLM reranker	0.9388	0.9390	0.9085
-	LLM reranker	0.9081	0.9390	0.9207
Others	vund	0.8970	-	-
-	UIT legalGrep	0.8970	-	-
-	NOWJ	0.8766	-	-
-	UTEHY-NLU	0.8482	-	-

Task 1: Evaluation on the private test

Evaluation Results (Task 2)

fit@hcmus

Teams	Accuracy	Rank
NOWJ	0.9878	1
Hallucineers	0.9878	1
UIT-Darkcow	0.9878	1
UTEHY-NLU	0.9756	4
Flame Reavers (Ours)	**0.9634**	**5**
vund	0.9634	5
BAMBOOVN	0.9512	7

Task 2: Evaluate on the private test

5. Conclusion

Conclusion

- Introduced a **dynamic hybrid architecture** combining sparse–dense retrieval and two-stage reranking.
- **Cross-Encoder** handles confident cases; **LLM** with “reason-and-extract” prompt tackles complex ones.
- Achieved **strong results**, validating the hybrid design.

Limitations

fit@hcmus

- **High Cost & Latency**: LLM components are computationally expensive and slow during inference.
- **Heavy LLM Dependency**: System performance is bottlenecked by the pre-trained LLM's accuracy and reasoning limits.
- **Static Reranking Threshold**: The fixed 0.8 confidence threshold was set empirically and may not be optimal for all cases.

Thank you very much for your attention